

2018. (A) 参考答案.

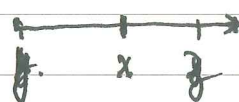
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Proof of (1)(a) $\vec{z}_n \xrightarrow{a.s.} \vec{z} \Leftrightarrow P\left(\bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} \{|\vec{z}_n - \vec{z}| \geq \varepsilon\}\right) = \lim_n P\left(\bigcup_{k=n}^{\infty} \{|\vec{z}_k - \vec{z}| \geq \varepsilon\}\right) = 0 \quad \forall \varepsilon > 0.$

$$\Rightarrow P\{\omega: |\vec{z}_n - \vec{z}| \geq \varepsilon\} \leq P\left(\bigcup_{k=n}^{\infty} \{|\vec{z}_k - \vec{z}| \geq \varepsilon\}\right) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

$$\Rightarrow \lim_n P\{\omega: |\vec{z}_n - \vec{z}| \geq \varepsilon\} = 0 \Rightarrow \vec{z}_n \xrightarrow{P} \vec{z}.$$

(b) suppose that $\vec{z}_n \xrightarrow{P} \vec{z}$.



$$\forall x \in C(F_{\vec{z}}), \quad \forall y < x \Rightarrow$$

$$\{\vec{z} < y\} = \{\vec{z} < y, \vec{z}_n < x\} + \{\vec{z} < y, \vec{z}_n \geq x\}$$

$$\subseteq \{\vec{z}_n < x\} \cup \{|\vec{z}_n - \vec{z}| \geq x - y\}$$

$$\Rightarrow F_{\vec{z}}(y) \leq F_{\vec{z}_n}(x) + P\{|\vec{z}_n - \vec{z}| \geq x - y\}$$

$$\Rightarrow F_{\vec{z}}(y) \leq \lim_{n \rightarrow \infty} F_{\vec{z}_n}(x) \quad \text{due to } \vec{z}_n \xrightarrow{P} \vec{z}.$$

$$\text{Similarly, } \lim_{n \rightarrow \infty} \overline{F}_{\vec{z}_n}(x) \leq \overline{F}_{\vec{z}}(x), \quad \forall x < \vec{z}.$$

$$\text{i.e., } F_{\vec{z}}(y) \leq \lim_{n \rightarrow \infty} F_{\vec{z}_n}(x) \leq \lim_n \overline{F}_{\vec{z}_n}(x) \leq \overline{F}_{\vec{z}}(x), \quad \forall y < x < \vec{z}$$

$$\text{As } y \uparrow x, \quad x \downarrow \vec{z}, \quad x \in C(F_{\vec{z}}), \Rightarrow$$

$$F_{\vec{z}}(x) \leq \lim_n F_{\vec{z}_n}(x) \leq \lim_{n \rightarrow \infty} \overline{F}_{\vec{z}_n}(x) \leq \overline{F}_{\vec{z}}(x)$$

$$\text{i.e., } \lim_n F_{\vec{z}_n}(x) = \overline{F}_{\vec{z}}(x), \quad \forall x \in C(F), \quad \text{i.e., } \vec{z}_n \xrightarrow{W} \vec{z}.$$

Proof of (2) (a) $\vec{z}_n \xrightarrow{P} \vec{z} \Leftarrow \vec{z}_n \xrightarrow{W} \vec{z}$. 不成立, 反例如下:

令 $\{\vec{z}, \vec{z}_n; n \geq 1\}$ 是 iid 随机列且 $\vec{z} \sim \begin{pmatrix} -1 \\ \frac{1}{2} \end{pmatrix}, \frac{1}{2}$. 则:

$$\forall \varepsilon > 0, \quad P\{|z_n - z| \geq \varepsilon\} = P\{z=1, z_n=-1\} + P\{z=-1, z_n=1\} \\ = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2} > 0$$

$$\text{i.e., } z_n \not\rightarrow z$$

$$\text{例. } F_{z_n}(x) = F_z(x), \Rightarrow \lim_n F_{z_n}(x) = F_z(x), \forall x \in \mathbb{R}$$

$$\text{i.e., } z_n \xrightarrow{w} z.$$

$$\Rightarrow z_n \xrightarrow{p} z \Leftarrow z_n \xrightarrow{w} z \text{ 不成立.}$$

$$(b) \quad z_n \xrightarrow{\text{a.s.}} z \Leftarrow z_n \xrightarrow{p} z \text{ 也不成立, 反例如下:}$$

$$\text{令 } \eta_{11}(w) = 1, \quad w \in \Omega = [0, 1),$$

$$\eta_{ki}(w) = \begin{cases} 1, & w \in [\frac{i-1}{k}, \frac{i}{k}), \\ 0, & \text{otherwise,} \end{cases} \quad i=1, 2, \dots, k.$$

$$\text{Define } z_1 = \eta_{11}, \quad z_2 = \eta_{21}, \quad z_3 = \eta_{22}, \quad z_4 = \eta_{31}, \dots$$

$$\text{则 } z_n \xrightarrow{\text{a.s.}} 0 = z, \quad \because \exists z_{n_{k_1}} = 1 \quad z_{n_{k_2}} = 0$$

$$\text{for large } k_1 \text{ and } k_2, \quad \lim_{k_1 \rightarrow \infty} z_{n_{k_1}} = 1, \quad \lim_{k_2 \rightarrow \infty} z_{n_{k_2}} = 0.$$

$$\text{But } \lim_n P\{|z_n| > \varepsilon\} = \lim_n P\left\{\left[\frac{i-1}{n}, \frac{i}{n}\right]\right\} = \lim_n \frac{1}{n} = 0$$

$$\text{i.e., } z_n \xrightarrow{p} 0 = z.$$

$$\text{i.e., } z_n \xrightarrow{p} z \not\Rightarrow z_n \xrightarrow{\text{a.s.}} z.$$

Proof of (3). $P\{w: |z_n \eta_n| \geq \varepsilon\} \leq P\{|z_n| \geq m\} + P\{|\eta_n| \geq \frac{\varepsilon}{m}\}$
 $\leq \sup_n P\{|z_n| \geq m\} + P\{|\eta_n| \geq \frac{\varepsilon}{m}\}$
 $\& \eta_n \xrightarrow{w} 0 \Leftrightarrow \eta_n \xrightarrow{p} 0$

$$\Rightarrow \lim_n P\{\omega: |\bar{z}_n \eta_n| \geq \varepsilon\} \leq \lim_{m \rightarrow \infty} \sup_n P\{|\bar{z}_n| \geq m\} + \lim_{m \rightarrow \infty} P\{|\eta_n| \geq \frac{\varepsilon}{m}\} = 0$$

$$\text{i.e., } \bar{z}_n \eta_n \xrightarrow{P} 0 \Leftrightarrow \bar{z}_n \eta_n \xrightarrow{W} 0 \quad \#$$

2. Proof. (1) Let $X = \bar{z}_1 - 3\bar{z}_2 + 4\bar{z}_3$

$$Y = \bar{z}_1 - 2\bar{z}_2 + \bar{z}_3$$

$$\text{Then } \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} 1 & -3 & 4 \\ 1 & -2 & 1 \end{pmatrix} \begin{pmatrix} \bar{z}_1 \\ \bar{z}_2 \\ \bar{z}_3 \end{pmatrix} \triangleq A \bar{z}, \quad \bar{z} = \begin{pmatrix} \bar{z}_1 \\ \bar{z}_2 \\ \bar{z}_3 \end{pmatrix} \sim N(\bar{a}, B)$$

$$\bar{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 6 & 8 \\ 2 & 8 & 11 \end{pmatrix}$$

$$\text{Hence } \begin{pmatrix} X \\ Y \end{pmatrix} \sim N(A\bar{a}, ABA^T) = N\left(\begin{pmatrix} 7 \\ 0 \end{pmatrix}, \begin{pmatrix} 44 & -6 \\ -6 & 1 \end{pmatrix}\right) \\ = N\left(\begin{pmatrix} 7 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & r\sigma_1\sigma_2 \\ r\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}\right)$$

$$\therefore E\{X|Y\} = a_1 + \frac{r\sigma_1}{\sigma_2^2} (-Y - a_2)$$

$$a_1 = 7, \quad \frac{r\sigma_1}{\sigma_2^2} = \frac{r\sigma_1\sigma_2}{\sigma_2^2} = \frac{-6}{1} = -6, \quad a_2 = 0$$

$$\therefore E\{X|Y\} = 7 - 6Y$$

$$\text{i.e., } E\{\bar{z}_1 - 3\bar{z}_2 + 4\bar{z}_3 | \bar{z}_1 - 2\bar{z}_2 + \bar{z}_3\} = 7 - 6(\bar{z}_1 - 2\bar{z}_2 + \bar{z}_3)$$

$$= -6\bar{z}_1 + 12\bar{z}_2 - 6\bar{z}_3 + 7 \quad \#$$

$$(2) \quad \hat{z} = \ln \frac{1}{W} = \bar{z}_n \left(\sum_{k=1}^{n-1} \ln \frac{1}{\bar{z}_k} \right) \triangleq XY, \quad \begin{matrix} X \sim U_{(0,1)} \\ Y \sim T(1, n) \end{matrix}$$

且 X 与 Y 相互独立, $\therefore P(Z < z) = \iint_{0 \leq xy < z} f_X(x) f_Y(y) dx dy$

$$f_X(x) \triangleq I_{(0,1)}(x), \quad f_Y(y) = \frac{1}{\Gamma(n-1)} y^{n-2} e^{-y} I_{(0,\infty)}(y)$$

$$P\{Z < z\} = \int_0^1 dx \int_0^{\frac{z}{x}} \frac{1}{\Gamma(n-1)} y^{n-2} e^{-y} dy$$

$$\therefore f_Z(z) = \frac{1}{\Gamma(n-1)} \int_0^1 \left(\frac{z}{x}\right)^{n-2} e^{-\frac{z}{x}} \cdot \frac{1}{x} dx$$

$$\therefore f_W(w) = f_Z\left(\ln \frac{1}{w}\right) \cdot \left|\frac{1}{w}\right| = \frac{1}{\Gamma(n-1)} \int_0^1 \left(\frac{\ln \frac{1}{w}}{x}\right)^{n-2} e^{-\frac{1}{x} \ln \frac{1}{w}} \cdot \frac{1}{x} dx$$

$$\left(= \frac{1}{\Gamma(n-1)} \left(\ln \frac{1}{w}\right)^{n-2} \int_0^1 \left(\frac{1}{x}\right)^{n-2} \left(\ln \frac{1}{w}\right)^{-\frac{1}{x}} dx\right)$$

$$= \frac{1}{\Gamma(n-1)} \left(\ln \frac{1}{w}\right)^{n-2} \frac{1}{w} \int_0^1 \left(\frac{1}{x}\right)^{n-1} \left(\frac{1}{w}\right)^{-\frac{1}{x}} dx$$

$$\therefore f_W(w) = \frac{1}{\Gamma(n-1)} \left(\ln \frac{1}{w}\right)^{n-2} \frac{1}{w} \int_0^1 \left(\frac{1}{x}\right)^{n-2} w^{\frac{1}{x}} dx$$

$$\therefore W_n \text{ 的概率分布密度函数为 } f_{W_n}(w) = \begin{cases} \frac{1}{\Gamma(n-1)} \left(\ln \frac{1}{w}\right)^{n-2} \frac{1}{w} \int_0^1 w^{\frac{1}{x}} \left(\frac{1}{x}\right)^{n-2} dx, & w > 0 \\ 0, & w \leq 0 \end{cases}$$

3. Proof of (1). $P\{T < y\} = 1 - P\{T \geq y\} = 1 - [P\{Z_1 > y\}]^n = P\{Z_n > y\}$

$$Z_k \sim \text{Exp}(\lambda_k), \quad P\{Z_k > y\} = e^{-\lambda_k y}, \quad y > 0$$

$$\therefore F_T(z) = 1 - e^{-(\lambda_1 + \dots + \lambda_n)z}, \quad \text{i.e., } T \sim \text{Exp}\left(\sum_{k=1}^n \lambda_k\right)$$

Proof of (2) $P\{Z_2 \geq Z_1\} = \int_0^\infty P\{Z_2 \geq Z_1 \mid Z_1 = y\} dF_{Z_1}(y)$

$$= \int_0^\infty P\{Z_2 > y\} dF_{Z_1}(y)$$

$$= \int_0^\infty e^{-\lambda_2 y} d(1 - e^{-\lambda_1 y}) = \lambda_1 \int_0^\infty e^{-(\lambda_1 + \lambda_2)y} dy$$

$$= \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

Proof of (3): $P\{Z_1 = \min\{Z_1, \dots, Z_n\}\} = P\{Z_1 = \min\{Z_1, \min\{Z_2, \dots, Z_n\}\}\}$

$$\eta \triangleq \min\{Z_2, \dots, Z_n\} \sim \text{Exp}\{\lambda_2 + \dots + \lambda_n\} \text{ b'c } Z_i \text{ i.i.d.}$$

$$\therefore P\{Z_1 = \min\{Z_1, \dots, Z_n\}\} = P\{Z_1 \leq \eta\} = \frac{\lambda_1}{\lambda_1 + (\lambda_2 + \dots + \lambda_n)} \text{ (by (2))}$$

$$\therefore P\{Z_1 = \min\{Z_1, \dots, Z_n\}\} = \frac{\lambda_1}{\lambda_1 + \dots + \lambda_n}$$

Proof of (4) $g(t) = E\{Z_1 \mid Z_1 \wedge Z_2 = t\}$

$$P\{Z_1 \wedge Z_2 \geq t\} = P\{Z_1 \geq t\} P\{Z_2 \geq t\} = e^{-(\lambda_1 + \lambda_2)t}$$

$$\Rightarrow \int_u^\infty g(t) dF_{Z_1 \wedge Z_2}(t) = \left(\int_u^\infty g(t) e^{-(\lambda_1 + \lambda_2)t} dt \right) (\lambda_1 + \lambda_2)$$

$$= \int_0^\infty E\{Z_1 \mid Z_1 \wedge Z_2 = t\} I_{[u, \infty)}(t) dF_{Z_1 \wedge Z_2}(t)$$

$$= E\{Z_1 I_{\{Z_1 \wedge Z_2 \geq u\}}\} = E\{Z_1 I_{\{Z_1 > u\}}\} E\{I_{\{Z_2 > u\}}\}$$

$$= \left(\int_u^\infty x e^{-\lambda_1 x} dx \right) e^{-\lambda_2 u}$$

$$\text{i.e. } (\lambda_1 + \lambda_2) \int_u^\infty g(x) e^{-(\lambda_1 + \lambda_2)x} dx = e^{-\lambda_2 u} \int_u^\infty x e^{-\lambda_1 x} dx$$

$$\text{for } u \geq 0: (\lambda_1 + \lambda_2) g(u) e^{-(\lambda_1 + \lambda_2)u} = -\lambda_2 e^{-\lambda_2 u} \int_u^\infty x e^{-\lambda_1 x} dx - e^{-\lambda_2 u} \cdot u e^{-\lambda_1 u}$$

$$\Rightarrow g(u) = \frac{\lambda_2 e^{-\lambda_2 u} \int_u^\infty x e^{-\lambda_1 x} dx}{\lambda_1 + \lambda_2} + \frac{u}{\lambda_1 + \lambda_2}$$

$$\therefore g(u) = \frac{\lambda_2}{\lambda_1 + \lambda_2} e^{-\lambda_2 u} \int_u^\infty x e^{-\lambda_1 x} dx + \frac{u}{\lambda_1 + \lambda_2}$$

$$= \frac{\lambda_2}{\lambda_1 + \lambda_2} e^{-\lambda_2 u} \frac{1}{\lambda_1} e^{-\lambda_1 u} \left[\frac{\lambda_1 + u}{\lambda_1} \right] + \frac{u}{\lambda_1 + \lambda_2}$$

$$= \frac{\lambda_2}{(\lambda_1 + \lambda_2) \lambda_1^2} e^{-(\lambda_1 + \lambda_2)u} \cdot (\lambda_1 + u) + \frac{u}{\lambda_1 + \lambda_2}$$

$$\therefore E\{Z_1 | Z_1 \wedge Z_2\} = \frac{1}{\lambda_1 + \lambda_2} \left[\frac{\lambda_2}{\lambda_1^2} (\lambda_1 + \min\{Z_1, Z_2\}) e^{-(\lambda_1 + \lambda_2) \min\{Z_1, Z_2\}} + \min\{Z_1, Z_2\} \right]$$

$$\Rightarrow E\{Z_1 | Z_1 \wedge Z_2\} = \frac{\lambda_2 (\lambda_1 + \min\{Z_1, Z_2\}) e^{-(\lambda_1 + \lambda_2) \min\{Z_1, Z_2\}}}{(\lambda_1 + \lambda_2) \lambda_1^2} + \frac{\min\{Z_1, Z_2\}}{\lambda_1 + \lambda_2} \quad \#$$

4. Proof of w. $\triangleq dF_X(x) = f_X(x) dx \quad a < x_m \quad \mathbb{R} \quad F_X(x_m) = \frac{1}{2}$

$$E\{|x-a| - |x-x_m|\} = \int_{\mathbb{R}} [|x-a| - |x-x_m|] dF_X(x)$$

$$= \int_{-\infty}^a (a-x_m) dF + \int_a^{x_m} [2x-a-x_m] dF + \int_{x_m}^{\infty} (x_m-a) dF$$

$$= (a-x_m) F(a) + F(x_m) (x_m-a) - F(a) (a-x_m) - 2 \int_a^{x_m} F(x) dx + (x_m-a) (1-\frac{1}{2})$$

$$= (F(x_m) + \frac{1}{2}) (x_m-a) - 2 \int_a^{x_m} F(x) dx = (x_m-a) - 2 \int_a^{x_m} F(x) dx$$

$$= 2 \int_a^{x_m} (x-a) dF_X(x),$$

□

$$\text{i.e., } E\{|X-a|\} = E\{|X-x_m|\} + 2 \int_a^{x_m} (x-a) dF_X(x), \quad (1)$$

Similarly, (1) holds for $a \geq x_m$.

Proof of (2): As $D(Y|G)(\omega) = E\{Y^2|G\}(\omega) + (E(Y|G))^2(\omega)$.

$$\text{we have } E(D(Y|G)) = E(Y^2) - E((E(Y|G))^2). \quad (2)$$

On the other hand,

$$\begin{aligned} D(E(Y|G)) &= E((E(Y|G))^2) - (E(E(Y|G)))^2 \\ &= E((E(Y|G))^2) - (EY)^2. \quad (3) \end{aligned}$$

$$(2) + (3) \Rightarrow D(Y) = D(E(Y|G)) + E(D(Y|G)). \quad \#$$

Proof of (3): Define $f(x) = \arctan(x)$, $x \in \mathbb{R}$. Then f is a strictly increasing, bounded funct, hence, $\exists f(\xi)$, $\eta f(\eta)$, $\exists f(\eta)$ and $\eta f(\xi)$ are integrable r.v's on $(\Omega, \mathcal{F}, \mathbb{P}(\cdot))$. As

$$E\{\exists f(\eta)\} = E\{E\{\exists f(\eta)|\eta\}\} = E\{f(\eta)E\{\exists|\eta\}\} = E\{\eta f(\eta)\},$$

$$\text{and, similarly, } E\{\eta f(\xi)\} = E\{E\{\eta f(\xi)|\xi\}\} = E\{\exists f(\xi)\},$$

$$\text{we have } E\{(3-\eta)(f(\xi)-f(\eta))\} = 0. \Rightarrow (3-\eta)(f(\xi)-f(\eta)) \xrightarrow{\text{a.s.}} 0$$

$$\Rightarrow \exists - \eta \xrightarrow{\text{a.s.}} 0.$$

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5. Proof. of 右边不等式: Let $A_n \triangleq \{\omega: \max_{1 \leq j \leq n} |S_j - ES_j| > \varepsilon\}$, then

$$A_n = \sum_{k=1}^n B_k, \text{ where } B_1 = A_1, B_k = A_k \bar{A}_{k-1} \cdots \bar{A}_1, k \geq 2$$

$$P(A_n) = \sum_{k=1}^n P(B_k) \leq \frac{1}{\varepsilon^2} \sum_{k=1}^n E\{(S_k - ES_k)^2 I_{B_k}^{(\omega)}\}$$

Noting that $S_n - ES_n [(S_n - S_k) - E(S_n - S_k)] \sim (S_k - ES_k) I_{B_k}$

相互独立, $E\{(S_k - ES_k)^2 I_{B_k}\} \leq E\left[\left[(S_n - S_k) - E(S_n - S_k)\right]^2 I_{B_k} + (S_k - ES_k)^2 I_{B_k}\right]$

$$= E\left[(S_n - S_k) - E(S_n - S_k) + S_k - ES_k\right]^2 I_{B_k}$$

so $P(A_n) \leq \frac{1}{\varepsilon^2} \sum_{k=1}^n E\left[(S_n - ES_n)^2 I_{B_k}\right]$

$$\stackrel{1}{=} \frac{1}{\varepsilon^2} E\left[(S_n - ES_n)^2 \sum_{k=1}^n I_{B_k}\right] = \frac{1}{\varepsilon^2} E\left[(S_n - ES_n)^2 I_{A_n}\right]$$

$$\leq \frac{1}{\varepsilon^2} E(S_n - ES_n)^2 = \frac{1}{\varepsilon^2} D(S_n). \quad \#$$

Proof of left-side: $|X_n| \leq C \Rightarrow |X_n - EX_n| \leq 2C$

$$|(S_k - ES_k) I_{B_k}| \leq |(S_{k-1} - ES_{k-1}) I_{B_k}| + |(X_k - EX_k) I_{B_k}|$$

$$\leq \varepsilon + 2C$$

$$\Rightarrow E\{(S_n - ES_n)^2 I_{A_n}\} = \sum_{k=1}^n E\{(S_n - ES_n)^2 I_{B_k}\}$$

$$= \sum_{k=1}^n E\{(S_k - ES_k)^2 I_{B_k} + (S_n - S_k - E(S_n - S_k))^2 I_{B_k}\}$$

$$\leq (\varepsilon + 2C)^2 \sum_{k=1}^n P(B_k) + \sum_{k=1}^n E\{(S_n - S_k - E(S_n - S_k))^2\} P(B_k)$$

$$\leq (\varepsilon + 2C)^2 \sum_{k=1}^n P(B_k) + \sum_{k=1}^n \sigma_k^2 \cdot \sum_{k=1}^n P(B_k) = \left[(\varepsilon + 2C)^2 + D(S_n)\right] P(A_n) \quad (19)$$

On the other hand,

$$\begin{aligned} E\{(S_n - ES_n)^2 I_{A_n}\} &= D(S_n) - E\{(S_n - ES_n)^2 I_{A_n^c}\} \\ &\geq D(S_n) - \varepsilon^2 P(A_n^c) \\ &= D(S_n) - \varepsilon^2 + \varepsilon^2 P(A_n) \quad (2*) \end{aligned}$$

so, by (1*) and (2*), we have

$$P(A_n) [(\varepsilon + 2C)^2 + D(S_n)] \geq D(S_n) - \varepsilon^2 + \varepsilon^2 P(A_n)$$

$$\Rightarrow \underline{P(A_n)} \geq \frac{D(S_n) - \varepsilon^2}{(\varepsilon + 2C)^2 + D(S_n) - \varepsilon^2}$$

$$= \frac{(\varepsilon + 2C)^2 + D(S_n) - \varepsilon^2 - (\varepsilon + 2C)^2}{(\varepsilon + 2C)^2 + D(S_n) - \varepsilon^2}$$

$$= 1 - \frac{(\varepsilon + 2C)^2}{(\varepsilon + 2C)^2 - \varepsilon^2 + D(S_n)} \geq 1 - \frac{(\varepsilon + 2C)^2}{D(S_n)} \quad \#$$

Proof of 6.(1). $P\left\{\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} \{Z_k \neq \eta_k\}\right\} \leq \sum_{k=n}^{\infty} P\{w: Z_k \neq \eta_k\}, \forall n \geq 1$

$$\Rightarrow P\left\{\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} \{w: Z_k \neq \eta_k\}\right\} = 0$$

Let $N \triangleq \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} \{w: Z_k \neq \eta_k\}$, $\tilde{\Omega} \triangleq \Omega - N$. Then $\forall w \in \tilde{\Omega}$

$$\exists n_0(w) \text{ st } \forall n \geq n_0(w), Z_n(w) = \eta_n(w). \Leftrightarrow$$

$$\sum_{k=1}^{\infty} (Z_k - \eta_k) = \sum_{k=1}^{n_0} (Z_k - \eta_k), \Rightarrow \sum_{n=1}^{\infty} (Z_n - \eta_n) \text{ 几乎处处收敛.}$$

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Define $S_n = \sum_{k=1}^n (\xi_k - E\xi_k)$, $n \geq 1$. By Kolmogorov's ineq.,

$$P\{\omega: \max_{M \leq k \leq N} |S_k - S_M| > \varepsilon\} \leq \frac{1}{\varepsilon^2} D(S_N - S_M) = \frac{1}{\varepsilon^2} \sum_{k=M+1}^N D(\xi_k)$$

$$\Rightarrow P\{\omega: \max_{k \geq M} |S_k - S_M| > \varepsilon\} \leq \frac{1}{\varepsilon^2} \sum_{k=M+1}^{\infty} D(\xi_k) \rightarrow 0$$

(as $M \rightarrow +\infty$).

Hence, $Y_M \triangleq \sup_{n, m \geq M} |S_n - S_m| \Rightarrow Y_M \downarrow$ as $M \uparrow$

$$\Rightarrow P\{\omega: Y_M > 2\varepsilon\} \leq 2 P\{\omega: \sup_{m \geq M} |S_m - S_M| > \varepsilon\} \rightarrow 0$$

$$P\{\omega: \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} \{Y_k \geq 2\varepsilon\}\} = 0 \iff Y_M \xrightarrow{\text{a.s.}} 0 \quad \text{as } M \rightarrow +\infty$$

$\Rightarrow \{S_n\}$ is a Cauchy sequence a.s., $\Rightarrow \lim_n S_n$ exists. a.s.

$$\iff \sum_{n=1}^{\infty} (\xi_n - E\xi_n) \text{ converges a.s.}$$

$$\wedge \sum_{k=1}^{\infty} [(\xi_k - E\xi_k) - (\eta_k - E\eta_k)] \text{ converges a.s.} \Rightarrow$$

$$\sum_{n=1}^{\infty} (\eta_n - E\eta_n) \text{ converges a.s.}$$

$$(2) \sum_{n=1}^{\infty} \frac{D(\eta_n)}{n^2} = \sum_{n=1}^{\infty} D\left(\frac{\eta_n}{n}\right) < +\infty \xrightarrow{\text{by (1)}} \sum_{n=1}^{\infty} \frac{\eta_n - E\eta_n}{n} \text{ converges a.s.}$$

so, by Kronecker's lemma, $\frac{1}{n} \sum_{k=1}^n (\eta_k - E\eta_k) \xrightarrow{\text{a.s.}} 0$. ~~η_k~~
($n \rightarrow \infty$)

$$\text{i.e., } \frac{1}{n} (S_n - ES_n) \xrightarrow{\text{a.s.}} 0 \quad (n \rightarrow \infty), \quad S_n \triangleq \sum_{k=1}^n \eta_k$$

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7. Proof of 1 / $3n$

by (1)

Proof of 1 $f_{n,m}(t) \stackrel{\Delta}{=} E\{e^{itZ_{n,m}}\}$, $\sigma_{n,m}^2 = E Z_{n,m}^2 \rightarrow 0, n \rightarrow \infty$

$$f_{n,m}(t) = 1 + it E(Z_{n,m}) - \frac{1}{2} \sigma_{n,m}^2 t^2 + o(\sigma_{n,m}^2)$$

$$= 1 - \frac{1}{2} \sigma_{n,m}^2 t^2 + o(\sigma_{n,m}^2) \quad \text{for large } n,$$

$$f_n(t) = E\{e^{itS_n}\} = \prod_{m=1}^n f_{n,m}(t) \rightarrow e^{-\frac{1}{2}\sigma^2 t^2} ?$$

$$|f_n(t) - e^{-\frac{1}{2}\sigma^2 t^2}| \leq \left| \prod_{m=1}^n f_{n,m}(t) - \prod_{k=1}^n \left(1 - \frac{1}{2}\sigma_{n,m}^2 t^2\right) \right|$$

$$+ \left| \prod_{k=1}^n \left(1 - \frac{1}{2}\sigma_{n,m}^2 t^2\right) - e^{-\frac{1}{2}\sum_{m=1}^n \sigma_{n,m}^2 t^2} \right| + \left| e^{-\frac{1}{2}\sum_{m=1}^n \sigma_{n,m}^2 t^2} - e^{-\frac{1}{2}\sigma^2 t^2} \right|$$

$$\stackrel{\Delta}{=} \square_{1n} + \square_{2n} + \square_{3n}$$

$$\square_{1n}(t) = \left| \prod_{m=1}^n f_{n,m}(t) - \prod_{k=1}^n \left(1 - \frac{1}{2}\sigma_{n,m}^2 t^2\right) \right|$$

$$\leq \sum_{m=1}^n |f_{n,m}(t) - (1 - \frac{1}{2}\sigma_{n,m}^2 t^2)|$$

$$= \sum_{k=1}^n |E[e^{itZ_{n,m}} - 1 + itZ_{n,m} + \frac{1}{2}(itZ_{n,m})^2]|$$

$$\leq \sum_{k=1}^n E \min \left\{ \frac{|tZ_{n,m}|^3}{3!}, \frac{|tZ_{n,m}|^2}{2!} \right\}$$

$$\leq \sum_{k=1}^n E \min \{ |tZ_{n,m}|^3, |tZ_{n,m}|^2 \}$$

$$\leq \sum_{k=1}^n E \{ |tZ_{n,m}|^3 I_{\{|Z_{n,m}| < \varepsilon\}} + |tZ_{n,m}|^2 I_{\{|Z_{n,m}| \geq \varepsilon\}} \}.$$

$$\leq |t|^3 \varepsilon \sum_{k=1}^n \sigma_k^2 + t^2 \sum_{m=1}^n E \{ Z_{n,m}^2 I_{\{|Z_{n,m}| \geq \varepsilon\}} \}$$

(3*)

$$\leq |t|^3 \varepsilon \sigma^2 + t^2 \sum_{m=1}^n E \{ Z_{n,m}^2 I_{\{|Z_{n,m}| \geq \varepsilon\}} \} \rightarrow 0 \quad (\text{as } \varepsilon \rightarrow 0, n \rightarrow \infty)$$

$$\Delta_{2n}(t) = \left| \prod_{m=1}^n \left(1 - \frac{1}{2} \sigma_{n,m}^2 t^2\right) - \prod_{m=1}^n e^{-\frac{1}{2} \sigma_{n,m}^2 t^2} \right|$$

$$\leq \sum_{m=1}^n \left| e^{-\frac{1}{2} \sigma_{n,m}^2 t^2} - \left(1 - \frac{1}{2} \sigma_{n,m}^2 t^2\right) \right|$$

$$\leq \sum_{m=1}^n \left| -\frac{1}{2} \sigma_{n,m}^2 t^2 \right|^2 = \frac{t^4}{4} \max_{1 \leq k \leq n} \sigma_{n,k}^2 \sum_{m=1}^n \sigma_{n,m}^2$$

$$\leq \left(\frac{1}{4} t^4 \sigma^2\right) \max_{1 \leq k \leq n} \sigma_{n,k}^2$$

$$\sigma_{n,m}^2 = E \{ \tilde{z}_{n,m}^2 \} \leq \varepsilon^2 + E \{ \tilde{z}_{n,m}^2 I_{\{|\tilde{z}_{n,m}| \geq \varepsilon\}} \}$$

$$\Rightarrow \max_{1 \leq m \leq n} \sigma_{n,m}^2 \leq \varepsilon^2 + \sum_{m=1}^n E \{ \tilde{z}_{n,m}^2 I_{\{|\tilde{z}_{n,m}| \geq \varepsilon\}} \}$$

$$\rightarrow 0 \quad (n \rightarrow \infty, \varepsilon \rightarrow 0)$$

$$\therefore \Delta_{2n}(t) \rightarrow 0 \quad (n \rightarrow +\infty) \quad (4^*)$$

$$\Delta_{3n}(t) = \left| e^{-\frac{1}{2} \sum_{m=1}^n \sigma_{n,m}^2 t^2} - e^{-\frac{1}{2} \sigma^2 t^2} \right| \rightarrow 0 \quad (\text{by (1)}) \quad (5^*)$$

$$\text{By (3*) - (5*), } f_n(t) \rightarrow e^{-\frac{1}{2} \sigma^2 t^2}$$

$$\text{i.e., } S_n \xrightarrow{d} Y \sim N(0, \sigma^2)$$

#

8. Proof. $\because S_n$ is $\mathcal{F}_n$ measurable $\therefore \sigma(S_n) \subset \mathcal{F}_n$

$$\therefore E\{Y_{n+1} | \mathcal{F}_n\} = E\{E\{Y_{n+1} | \sigma(S_n)\} | \mathcal{F}_n\}$$

$$E\{Y_{n+1}$$

8. Proof. 令 $z = \frac{1}{n} \sum_{j=1}^n z_j$, $P_j(A) = \int I_A(\omega) z_j d\mu$, 则 P_j 为 $(\Omega, \mathcal{F})$ 的概率测度, 且 $\frac{1}{n} \sum_{j=1}^n E\{z_j z_j\} = E\{z z_j\} = E^{P_i}\{z\}$.

By Jensen 不等式: $\ln E^{P_i}\{z\} \geq E^{P_i}\{\ln z\}$. 则

$$\begin{aligned} \prod_{i=1}^n \left(\frac{1}{n} \sum_{j=1}^n E\{z_j z_j\} \right) &= \prod_{i=1}^n E\{z z_j\} = \prod_{i=1}^n E^{P_i}\{z\} \\ &= e^{\sum_{i=1}^n \ln E^{P_i}\{z\}} \geq e^{\sum_{i=1}^n E^{P_i}\{\ln z\}} \end{aligned}$$

$\sum_{i=1}^n \ln z$

$\ln z$

$$\begin{aligned} \sum_{i=1}^n E^{P_i}\{\ln z\} &= n E^P\{z \log z\} \quad (x \ln x \geq x - 1, x > 0) \\ &\geq n E^P\{z - 1\} = n [E^P\{z\} - 1] = n[1 - 1] = 0 \end{aligned}$$

$$\begin{aligned} \therefore \prod_{i=1}^n \left(\frac{1}{n} \sum_{j=1}^n E\{z_j z_j\} \right) &\geq e^{\sum_{i=1}^n E^{P_i}\{\ln z\}} = e^{n E^P\{z \ln z\}} \\ &\geq e^{n E^P\{z - 1\}} = e^{n[1 - 1]} = e^0 = 1. \end{aligned}$$

$$\text{i.e., } \prod_{i=1}^n \left(\frac{1}{n} \sum_{j=1}^n E\{z_j z_j\} \right) \geq 1.$$

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9. Proof. To show that $E\{Y_{n+1} | \mathcal{F}_n\} = Y_n$, we need to show

$\forall$ bounded borel($\mathbb{R}$)-measurable $F(x_1, \dots, x_n)$,

$$E\{(Y_{n+1} - Y_n) F(z_1, \dots, z_n)\} = 0, \text{ where } Y_n = \frac{1}{\sqrt{n+1}} \exp\left\{\frac{S_n^2}{2(n+1)}\right\}$$

$\forall (x_1, \dots, x_n)$, define $S_n = \sum_{k=1}^n x_k$. Then.

$$\square \triangleq E\{(Y_{n+1} - Y_n) F(z_1, \dots, z_n)\} = \int_{\mathbb{R}^n} E\{(Y_{n+1} - Y_n) F(z_1, \dots, z_n) \mid z_1 = x_1, \dots, z_n = x_n\} dF_{z_1}(x_1) \dots dF_{z_n}(x_n)$$

$$= \int_{\mathbb{R}^n} E\left\{\left(\frac{1}{\sqrt{n+2}} \exp\left\{\frac{(S_n + z_{n+1})^2}{2(n+2)}\right\} - \frac{1}{\sqrt{n+1}} \exp\left\{\frac{S_n^2}{2(n+1)}\right\}\right) F(x_1, \dots, x_n)\right\} dF_{z_1}(x_1) \dots dF_{z_n}(x_n)$$

$$\text{Hence: } E\left\{\frac{1}{\sqrt{n+2}} \exp\left\{\frac{(S_n + z_{n+1})^2}{2(n+2)}\right\} F(x_1, \dots, x_n)\right\} \quad S_n = \sum_{k=1}^n x_k.$$

$$= F(x_1, \dots, x_n) \frac{1}{\sqrt{n+2}} E\left\{\exp\left\{\frac{(z_{n+1} + S_n)^2}{2(n+2)}\right\}\right\}$$

$$\frac{1}{\sqrt{n+2}} E\left\{\exp\left\{\frac{(z_{n+1} + S_n)^2}{2(n+2)}\right\}\right\} = \frac{1}{\sqrt{n+2}} \int_{\mathbb{R}} e^{\frac{(x + S_n)^2}{2(n+2)}} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

$$= \frac{1}{\sqrt{n+2}} \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{\left[-\frac{(x - \frac{S_n}{n+1})^2}{2\sigma^2} + \frac{(n+2)}{2(n+1)^2} S_n^2\right]} dx$$

$$= \frac{1}{\sqrt{n+2}} \frac{1}{\sqrt{2\pi}} e^{\frac{n+2}{2(n+1)^2} S_n^2} \int_{\mathbb{R}} e^{-\frac{(x - \frac{S_n}{n+1})^2}{2\sigma^2}} dx, \quad \sigma^2 = \frac{n+2}{n+1}$$

$$= \frac{1}{\sqrt{n+2}} \frac{1}{\sqrt{2\pi}} e^{\frac{S_n^2}{2(n+1)}} \sqrt{2\pi} \cdot \sigma = \frac{1}{\sqrt{n+1}} e^{\frac{S_n^2}{2(n+1)}}$$

$$\text{So } \square = \int_{\mathbb{R}^n} E\{F(x_1, \dots, x_n) \left[\frac{1}{\sqrt{n+1}} e^{\frac{S_n^2}{2(n+1)}} - \frac{1}{\sqrt{n+1}} e^{\frac{S_n^2}{2(n+1)}} \right]\} dF_{z_1}(x_1) \dots dF_{z_n}(x_n)$$

= 0

i.e., $E\{Y_{n+1} | \mathcal{F}_n\} \stackrel{a.s.}{=} Y_n$. i.e., $\{Y_n, \mathcal{F}_n, n \geq 1\}$ is a martingale.